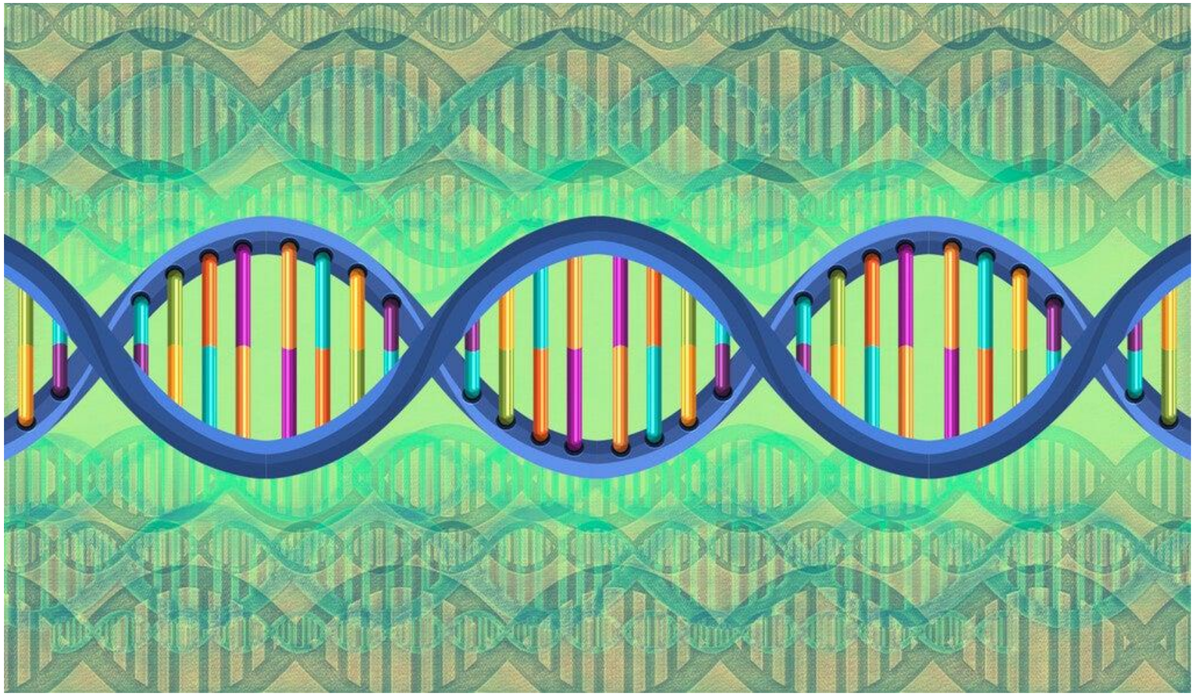
‘Jumping’ genes yield new clues to origins of neurodegenerative disease

Massive, repetitive stretches of DNA in the human genome may harbor hints about the onset of a rare, inherited neurodegenerative disorder, a new study finds.



Massive, repetitive stretches of DNA in the human genome may harbor hints about the onset of a rare, inherited neurodegenerative disorder called ataxia-telangiectasia as well as other related diseases, a new Yale School of Medicine study finds. The findings also point to possible directions for treatment of the disease.

The results are published Sept. 6 in the journal Neuron.

[‘Jumping’ genes yield new clues to origins of neurodegenerative disease | YaleNews]

The Dynamic Genome

- Discovery of transposable elements through study of “unstable” alleles
- Impact of transposable elements on the genome
- Genomic surveillance mechanisms that limit transposon mobility

Discovery of transposable elements through study of “unstable” alleles

Careful observation of "unstable" pigment gene alleles led to the discovery of transposable elements

Experiment: use pollen from a C^1 (dominant negative allele) strain to fertilize C/C ovules

<u>Ovule</u>

C/C

X

<u>Pollen</u>

C^1/C^1

C^1/C

Predicted result:
all colorless
kernels

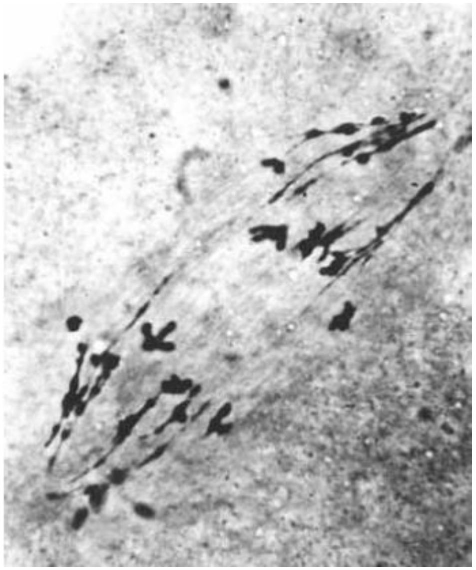
The order of dominance is $C^I > C > c$.

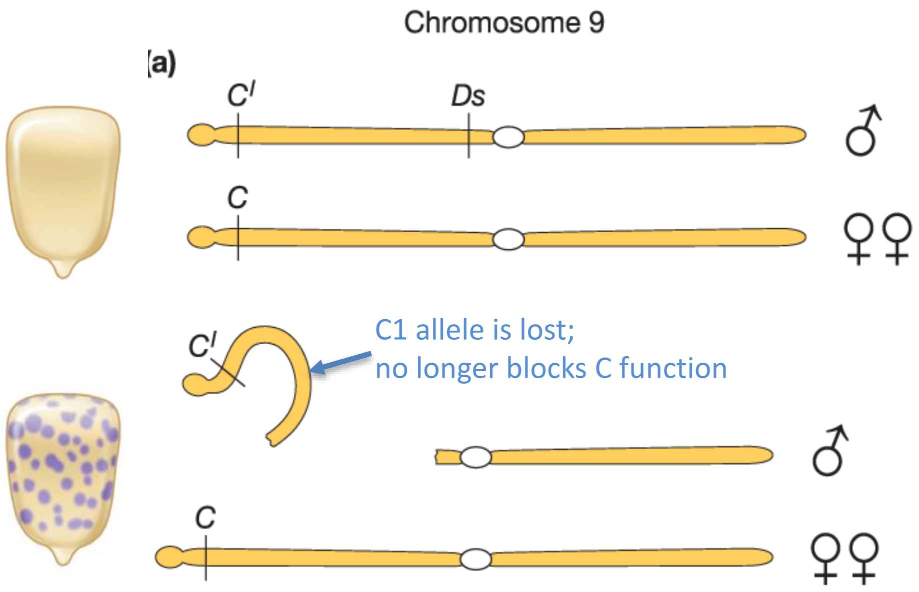
unexpected result from one particular "pollen parent":
some kernels have blue pigment! In sectors!
How can this be?

Careful observation of "unstable" pigment gene alleles led to the discovery of transposable elements

Hypothesis 1: C1 allele is lost in some cells of the developing endosperm
("if these cells can produce pigment, then maybe C1 is no longer inhibiting pigment production, e.g. maybe it's just not there")

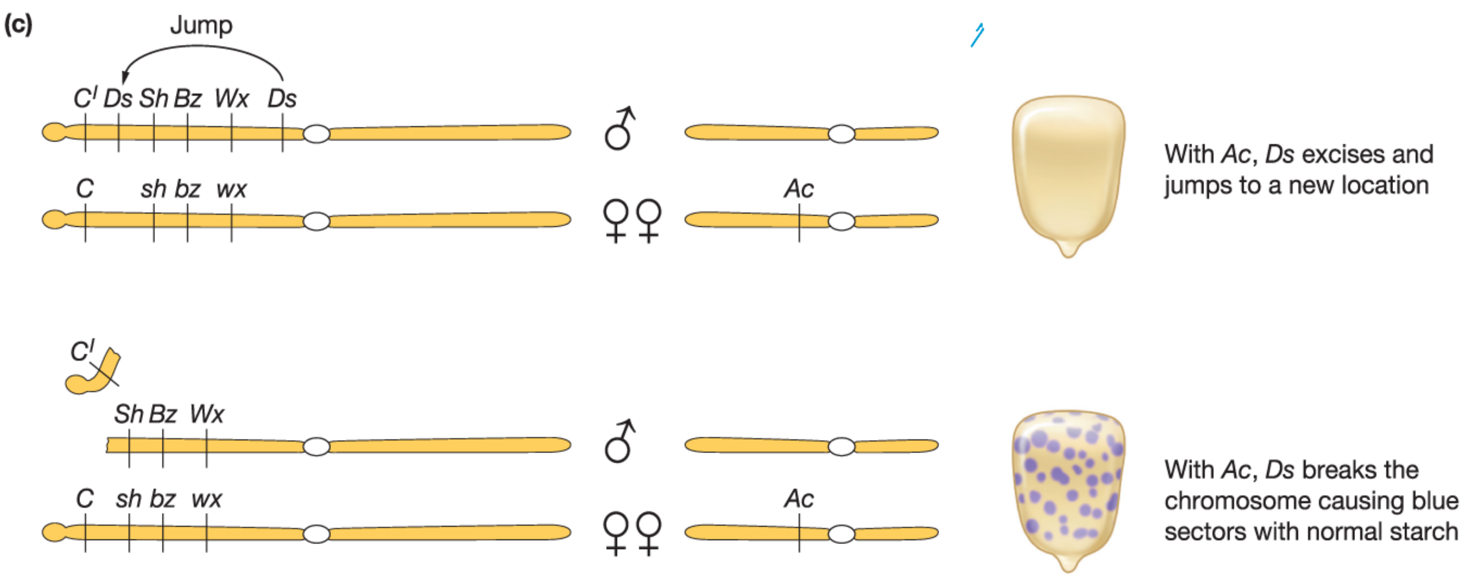
Experiment: look for breakage in chromosomes of the pollen parent
Result: high rate of breakage in this strain





Interpretation: loss of the C1 allele is caused by chromosome breakage.

Evidence for <u>movement</u> of these genetic elements: Breaks occur in a different location in a related strain

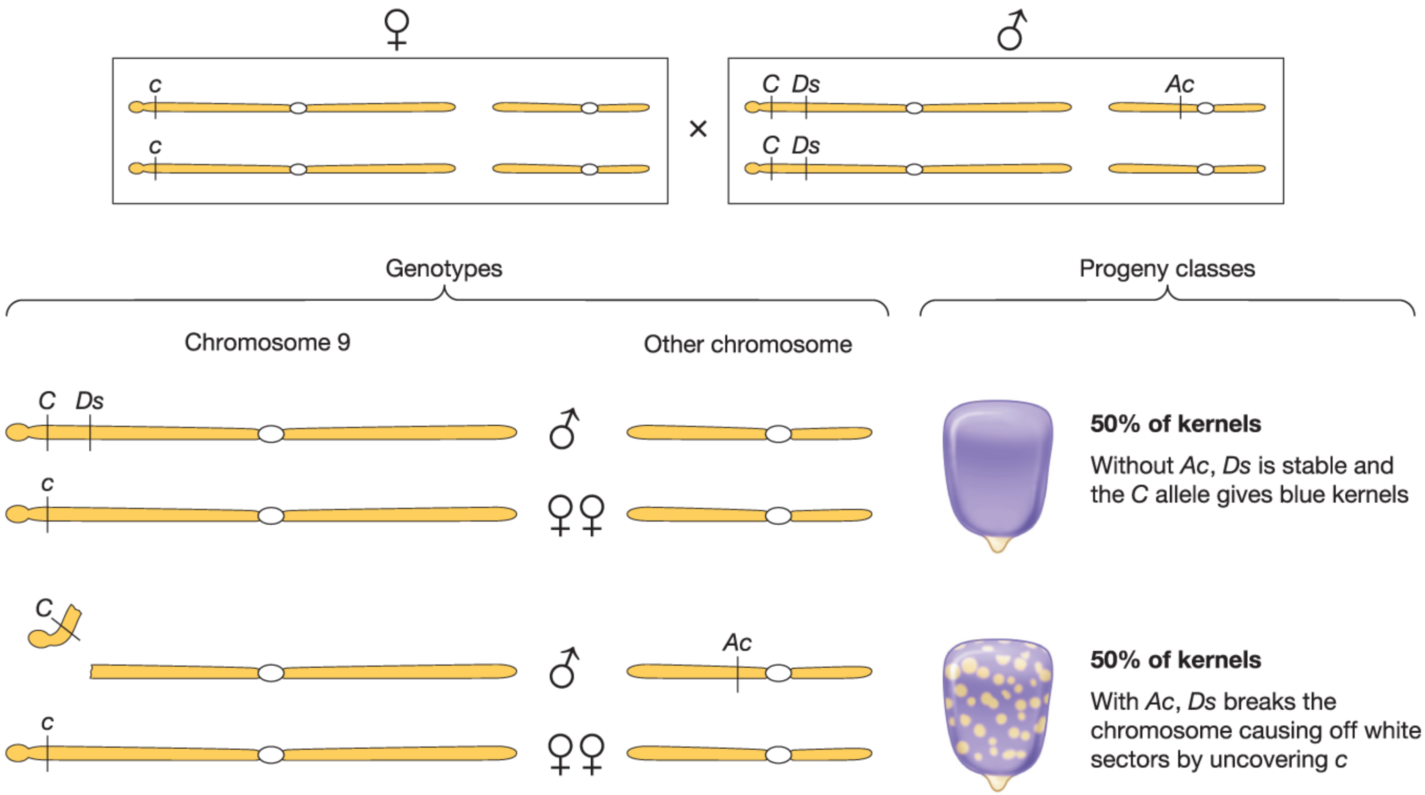


Interpretation: In some strains, Ds has moved to a new location. This new location can be inferred from the phenotype of the sectors.

Observation: in these sectors, pigment is restored but other kernel traits (Sh, Bz, Wx) remain unchanged

Note:
Not all Ds element insertion sites are associated with chromosome breaks!

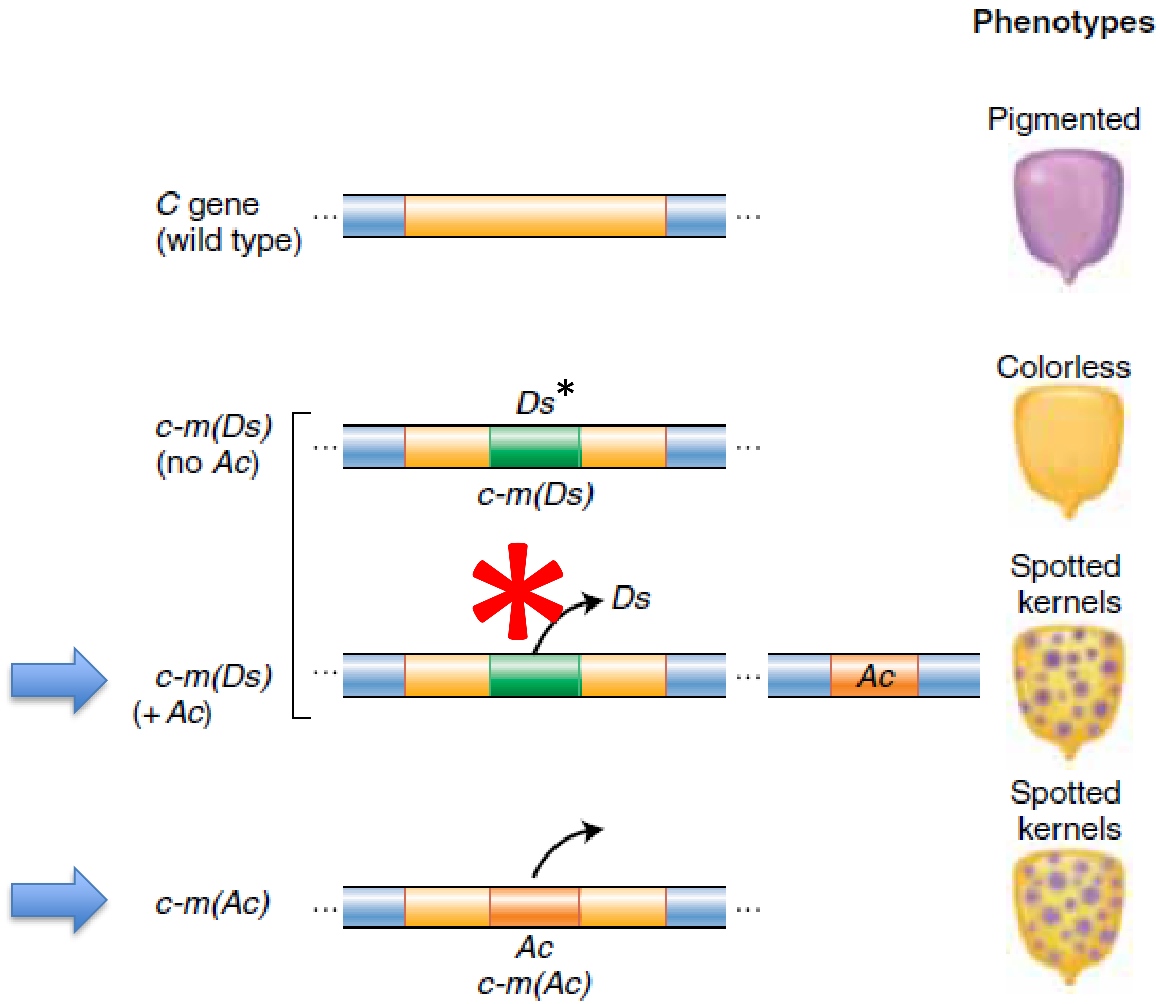
Evidence for movement of these genetic elements: multiple "jumps", and inability to map Ac element



Transposable elements can be non-autonomous (like Ds) or autonomous (like Ac)

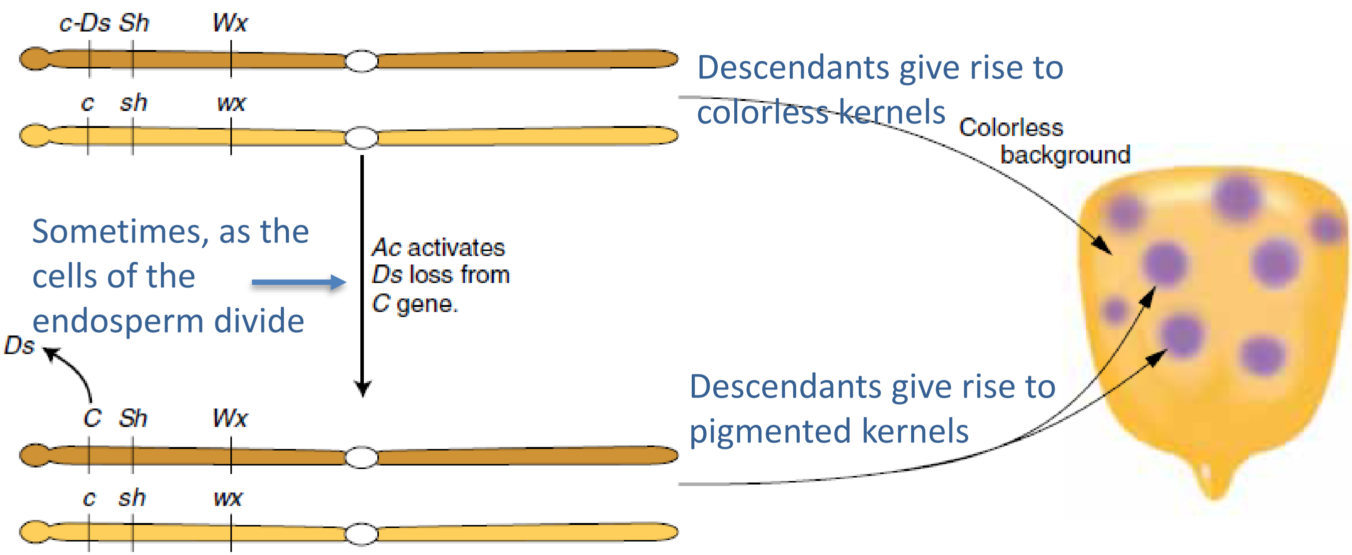
Requires *Ac* to "jump out" (aka "excise" or "transpose")
"non-autonomous"

Ac jumps out on its own
"autonomous"

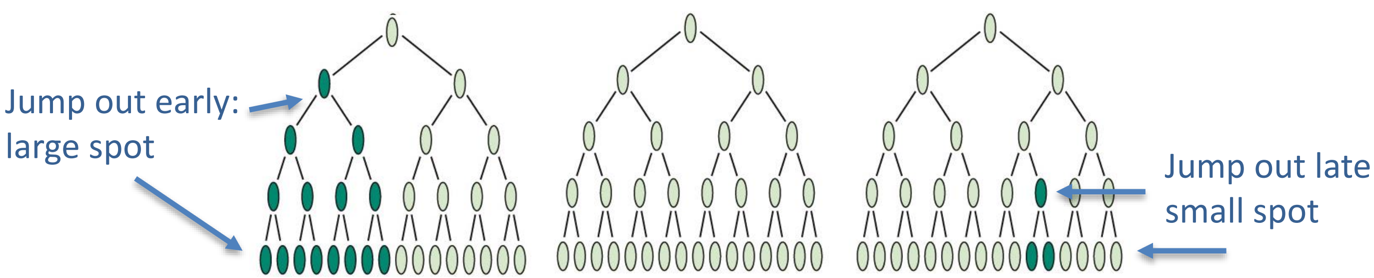


**The type of Ds is found in the c-Ds allele has a different structure. When the Ds in c-Ds excises, the two ends of the chromosome are joined together, restoring a functional dominant C allele.*

Spot size depends on when, during endosperm development, the C gene became active



Spot size depends on when in development the Ds element jumped out and restored C function.



Transposable elements

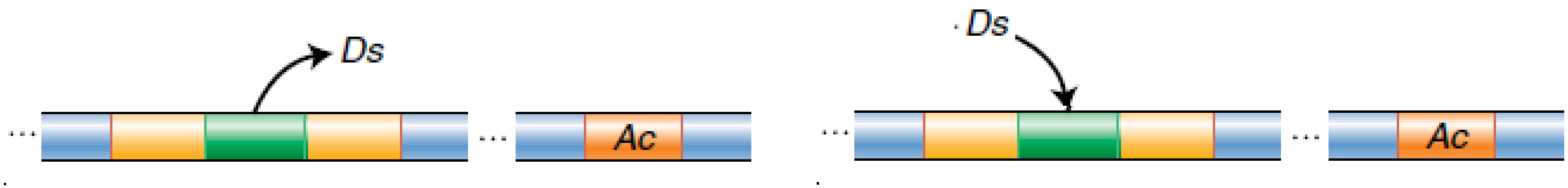
McClintock had discovered transposons, also known as "transposable elements".

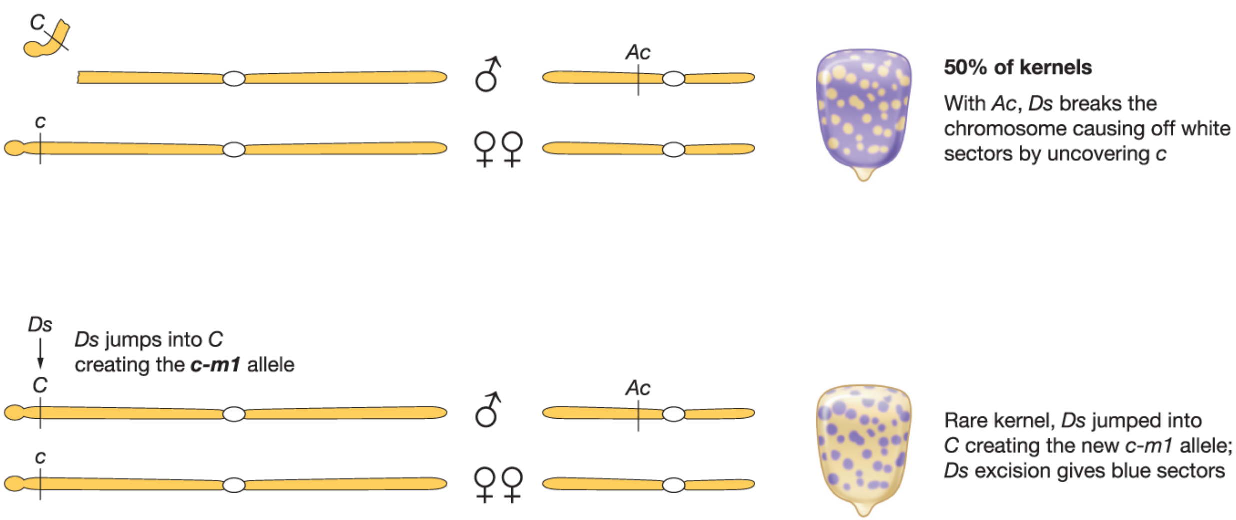
"From examination of instability of genic action at a number of known loci in maize, it is concluded that mutations need not express changes in genes, but may be the result of changes affecting the control of genic action."

Barbara McClintock 1953

Think Break

In the presence of Ac, Ds mobilization reverts the c(Ds) mutant allele frequently. But in rare cases Ds mutates the C allele. What happens, and why the difference in frequency?





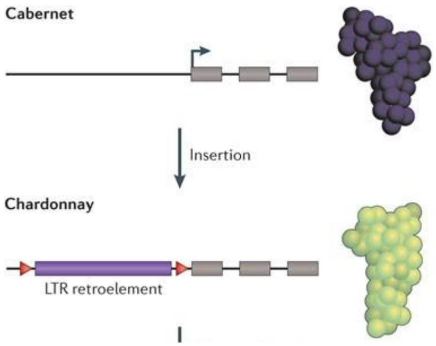
Jumping out: frequent

Jumping in: rare

Griffiths et al., *Introduction to Genetic Analysis*, 12e, © 2020 W. H. Freeman and Company

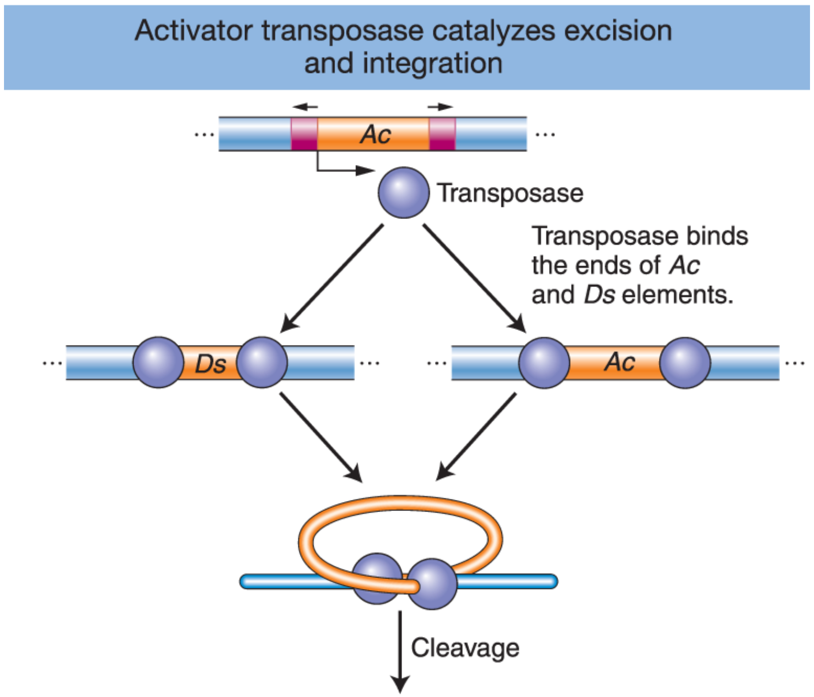
Transposable elements are found in all organisms

- Transposable elements are everywhere. Not just a weird plant phenomenon!
- Studies in maize identified transposable elements genetically, but it was later studies in yeast and Drosophila that identified the actual pieces of DNA that are mobile and discovered how they move.
- Transposons, or “transposable elements” are ancient, found in all organisms: bacteria, plants, yeast, multicellular animals.
- Transposable elements fall into two classes:
  - Class 1: Retrotransposons (eukaryotes)
  - Class 2: DNA transposons (prokaryotes and eukaryotes)

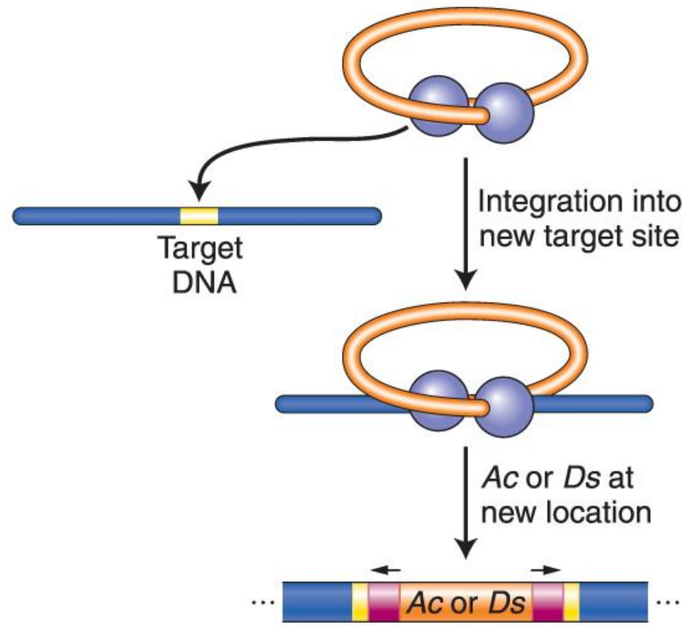


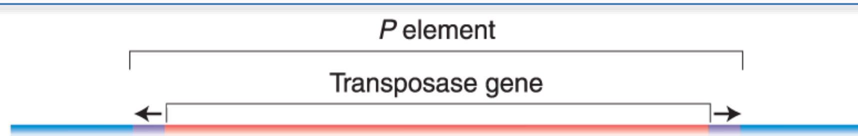
Class 2: DNA transposons

Example: the Ds and Ac elements in maize



There are multiple transposon families in maize, each with autonomous and non-autonomous elements. *Ac* and *Ds* are in the same family. Each has a transposase version that is specific for its own family.





Another example: P-elements in Drosophila

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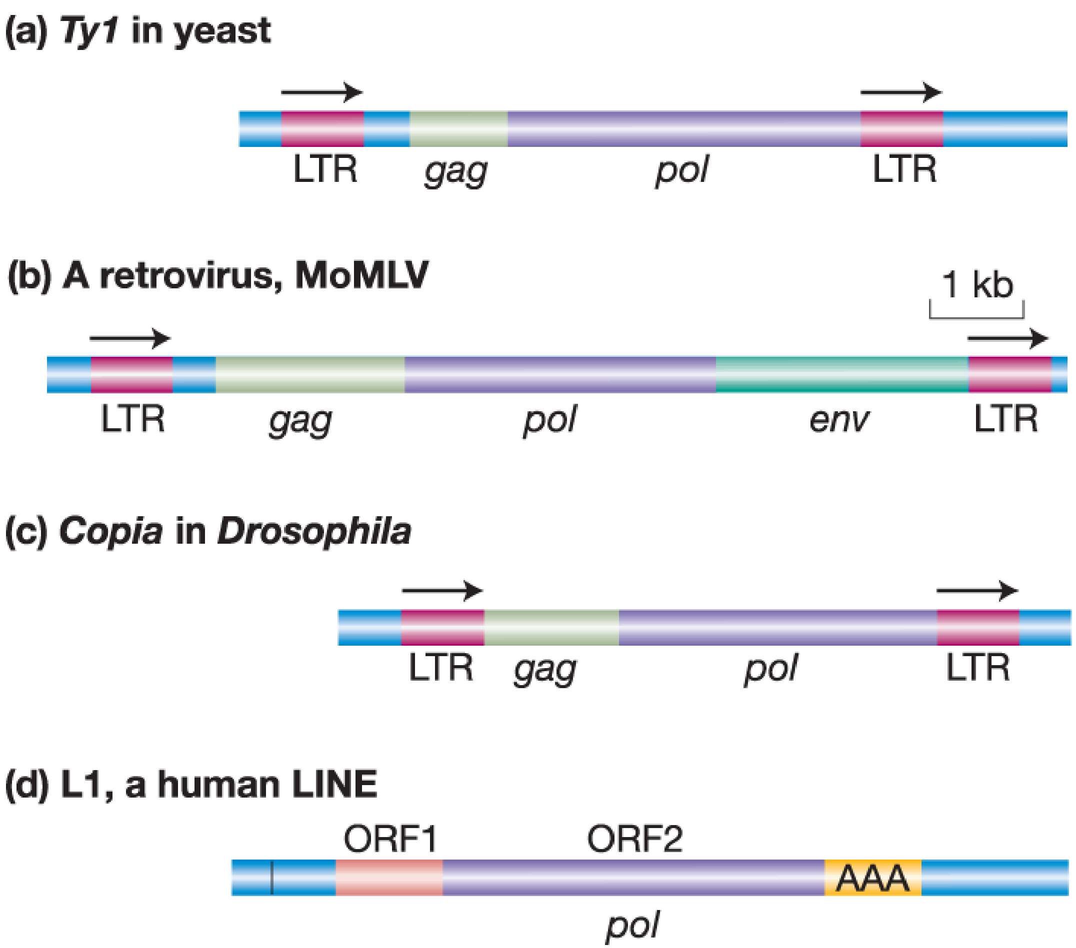
Class 1: Retrotransposons



- Similar to retroviruses
- First identified in yeast as a repeated sequence in the genome.
- Found in many organisms
- LINE (long interspersed element) is a retrotransposon found in the human genome.

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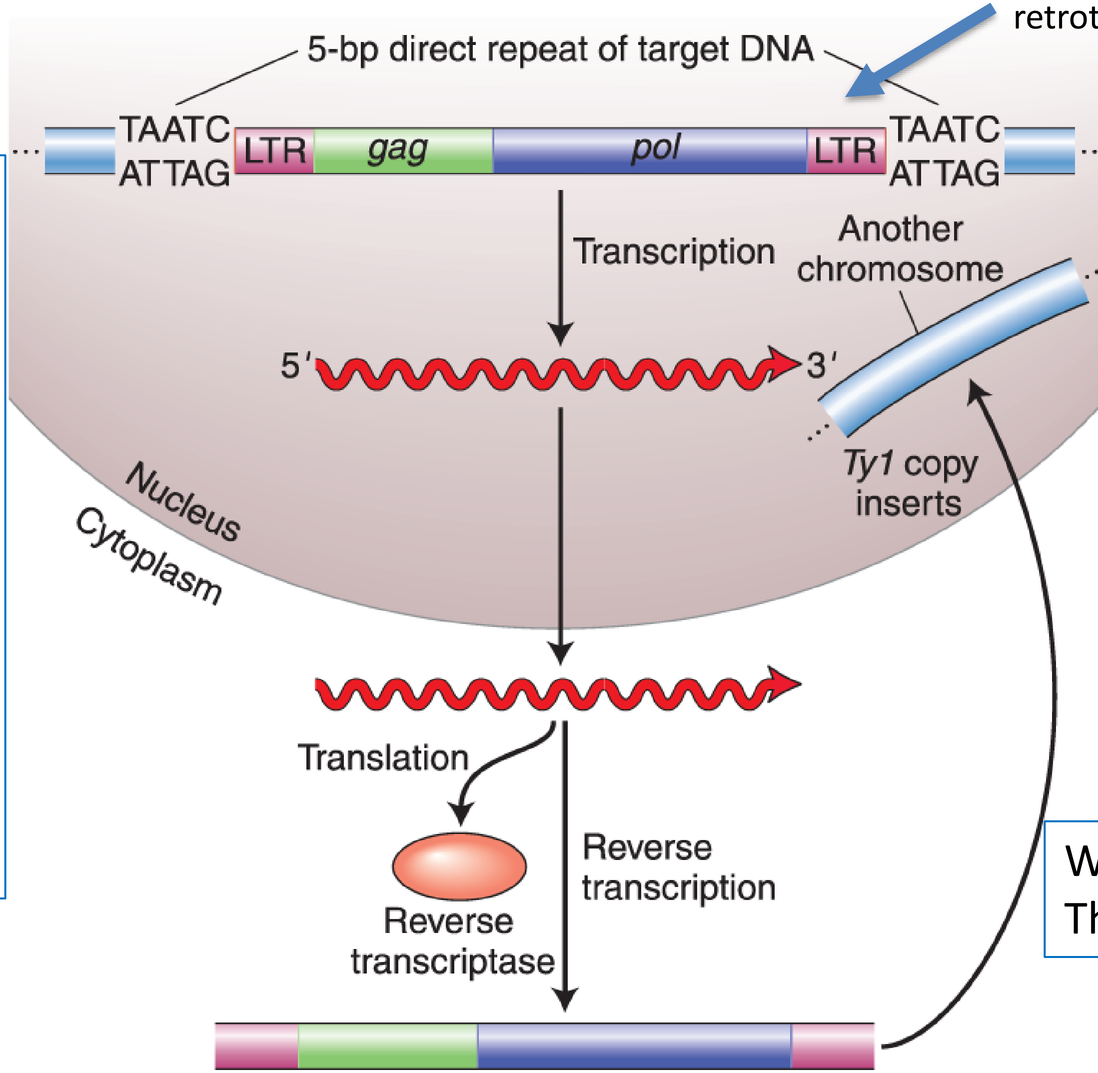
Retrotransposons move through an RNA intermediate

Example: yeast Ty1 retrotransposon



- Retrotransposons move via an RNA intermediate.
- Transposition is mediated by reverse transcriptase (encoded by the retrotransposon *pol* gene), not transposase.
- This is a replicative ("copy-paste") mechanism.

What's missing? The env gene.

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What is the impact of these transposable elements on the genome?

Transposable elements are abundant in large genomes

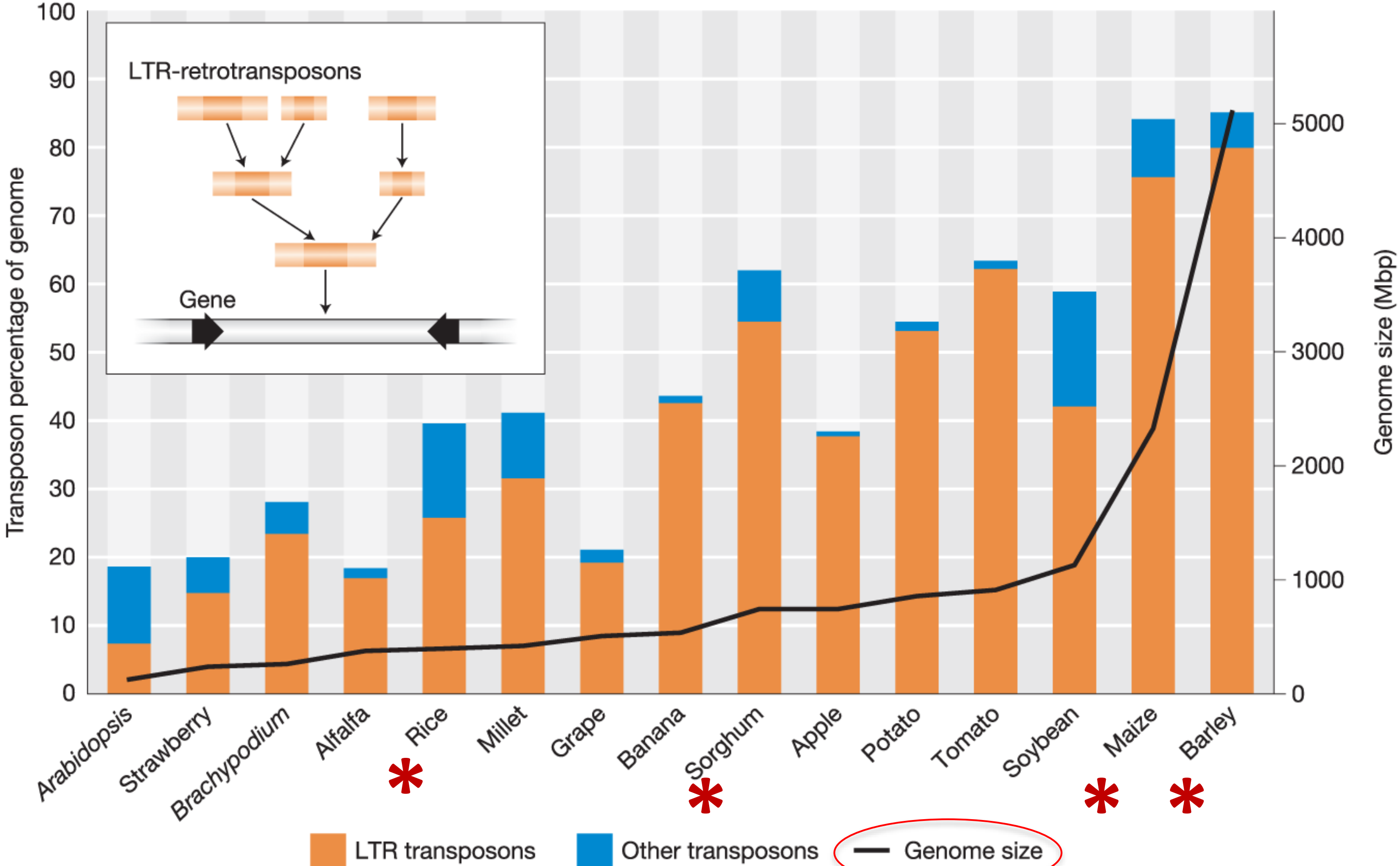
Transposable elements in the human genome

Element	Transposition	Structure	Length	Copy number	Fraction of genome
LINEs Class 1	Autonomous	ORF1 ORF2 *(pol)*  AAA	1–5 kb	20,000–40,000	21%
SINEs Class 1	Nonautonomous	ex: Alu  AAA	100–300 bp	1,500,000	13%
DNA transposons Class 2	Autonomous	← transposase → 	2–3 kb	300,000	3%
	Nonautonomous	← → 	80–3000 bp		

Figure 15-21
Introduction to Genetic Analysis, Eleventh Edition

There is about 20 times as much DNA in the human genome derived from transposable elements as there is DNA encoding all human proteins.

Transposable elements in plants account for differences in genome size



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Transposable elements in plants account for differences in genome size

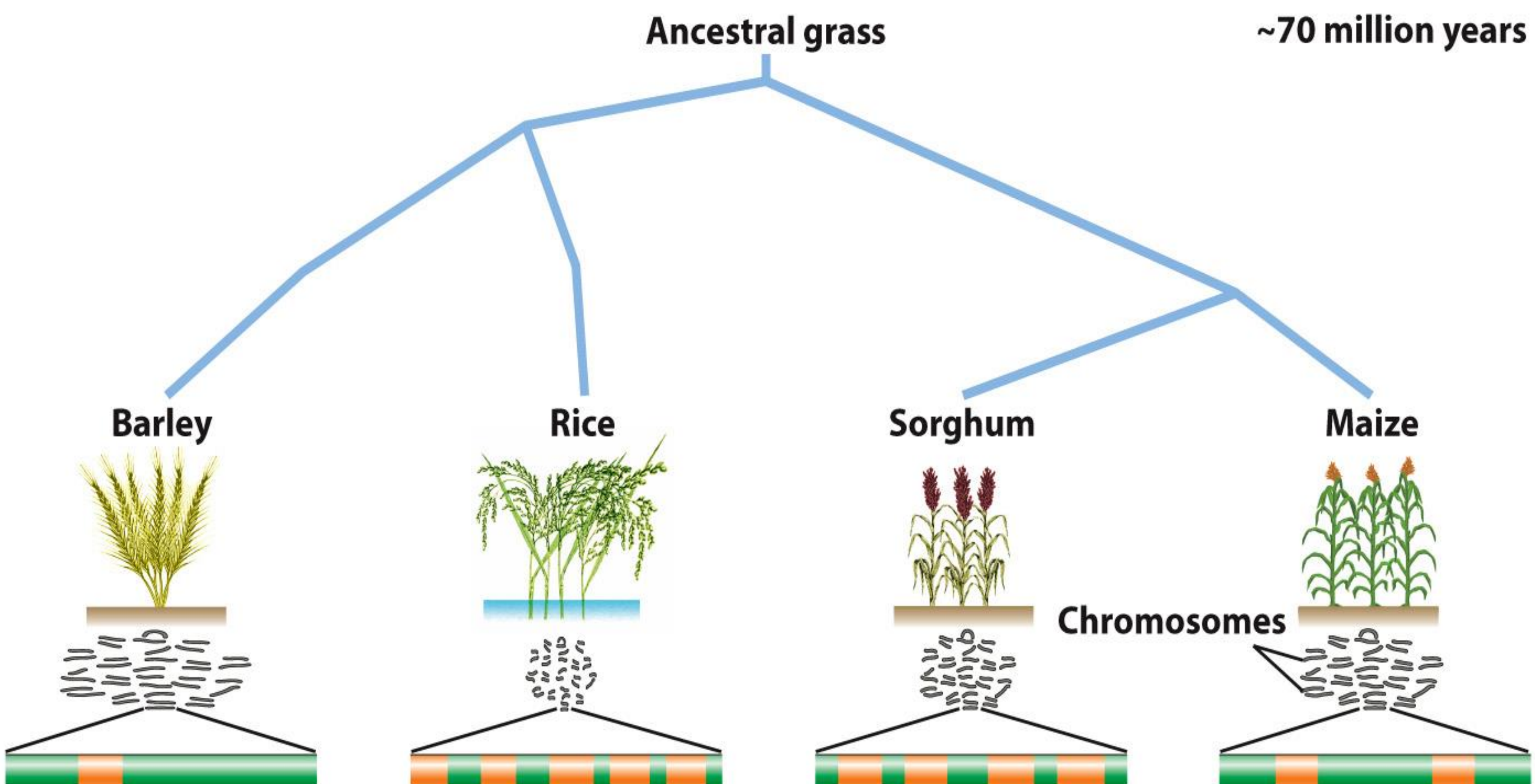
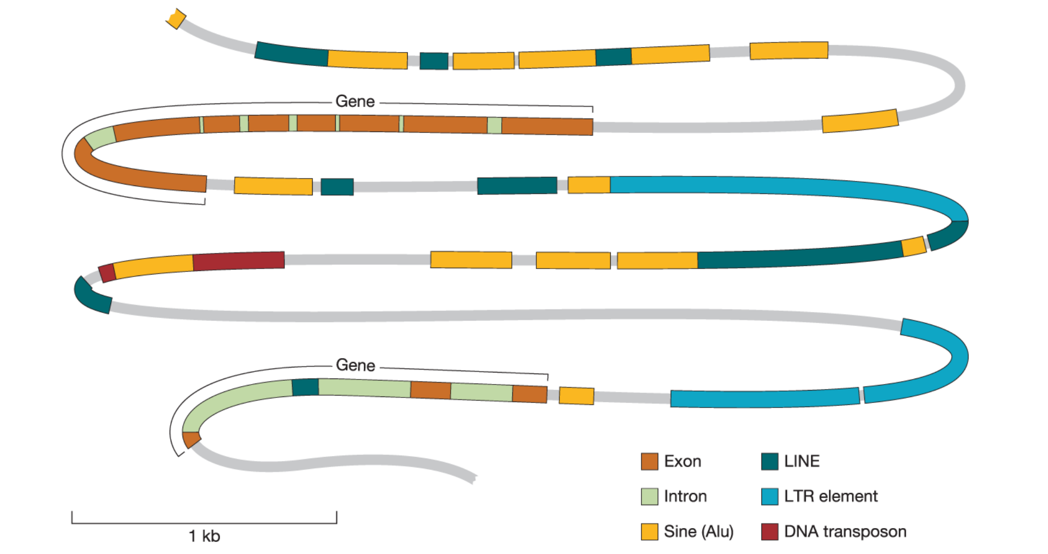


Figure 15-23
Introduction to Genetic Analysis, Eleventh Edition

Genes (orange) TEs (green)

Transposable elements are usually found in introns or intergenic regions



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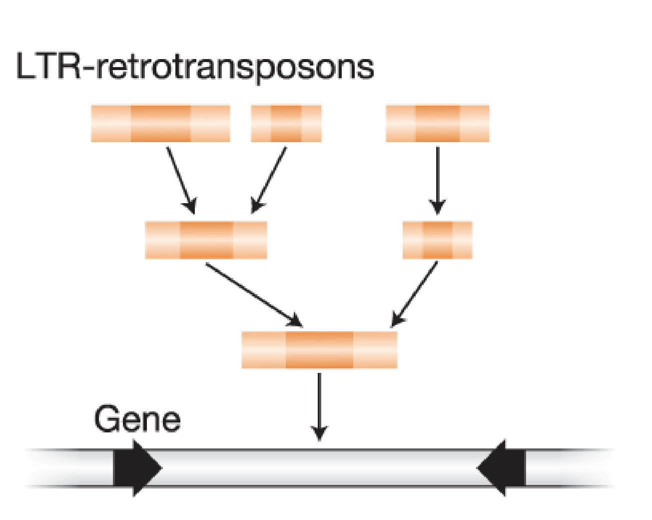
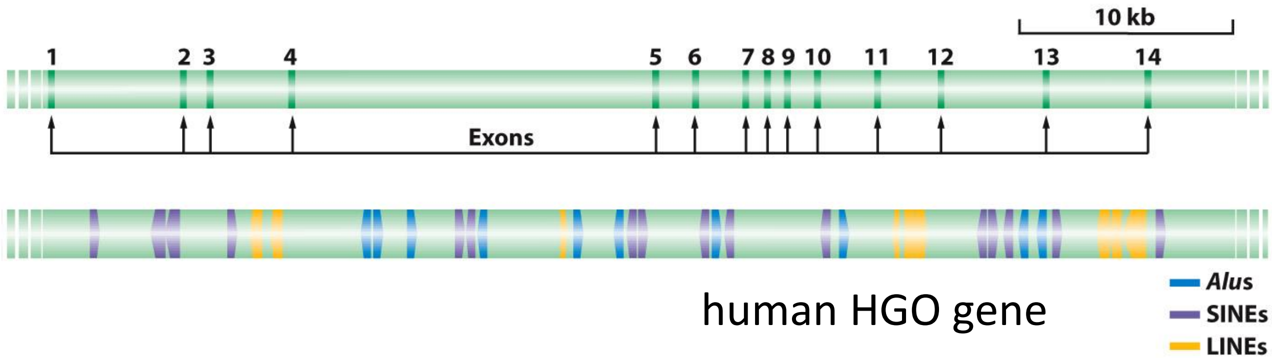




human HGO gene

Transposable elements can affect genes

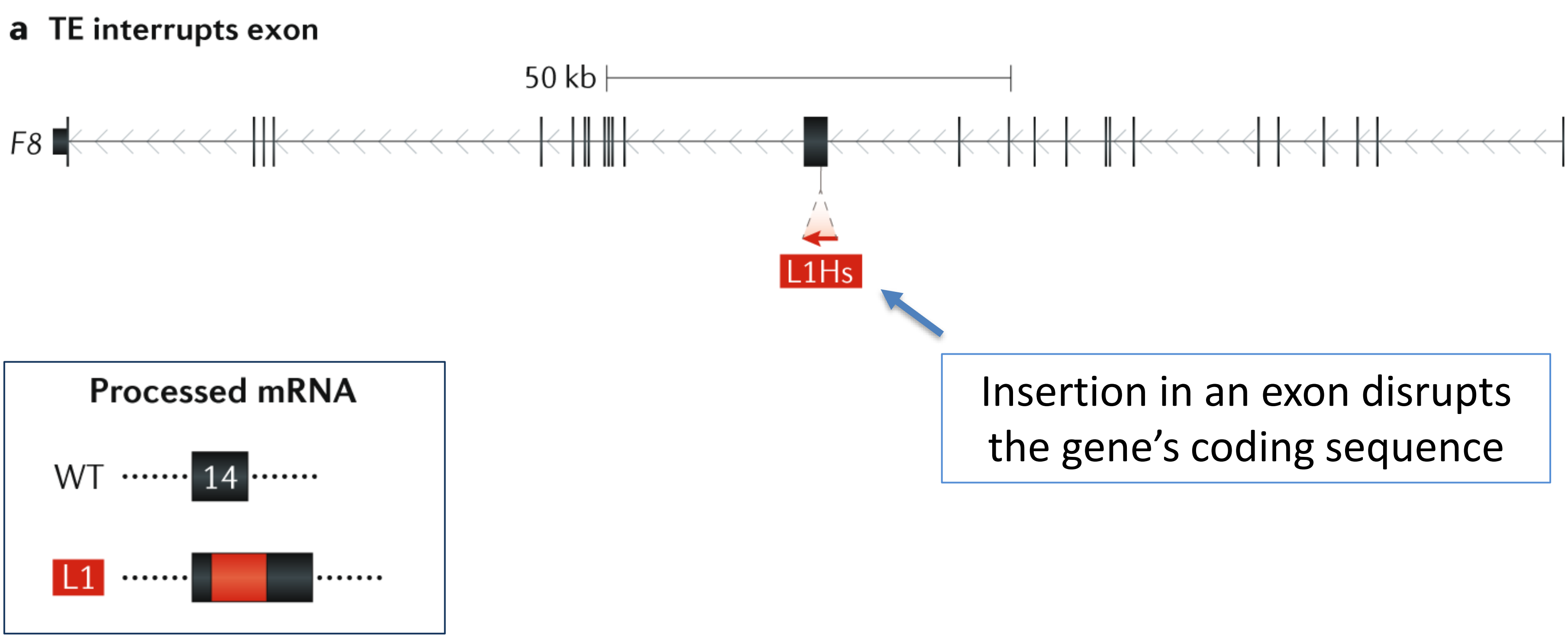
Element	Transposition	Structure	Length	Copy number	Fraction of genome
LINEs	Autonomous	ORF1 ORF2 *(pol)* AAA	1–5 kb	20,000–40,000	21%
SINEs	Nonautonomous	AAA	100–300 bp	1,500,000	13%
DNA transposons	Autonomous	← transposase →	2–3 kb	300,000	3%
	Nonautonomous	← →	80–3000 bp		

Figure 15-21
Introduction to Genetic Analysis, Eleventh Edition

1/600 spontaneous mutations causing important human diseases results from the transposition of a LINE or SINE element

Transposable elements can affect genes

Example: *Homo sapiens*-specific LINE-1 (L1Hs) insertion in a patient with haemophilia A (a blood clotting disorder) interrupts a coding exon of factor VIII (F8)

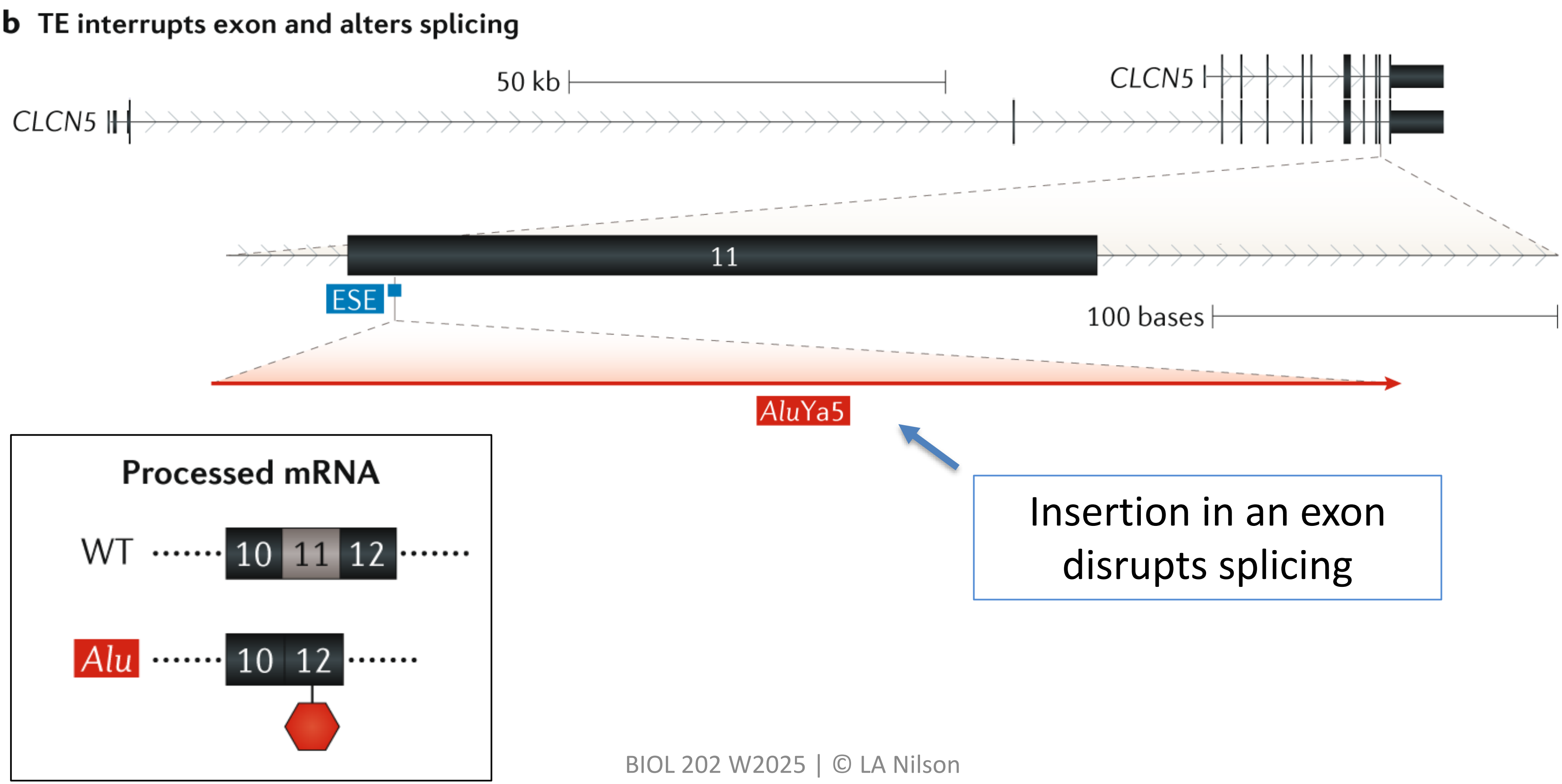


Insertion in an exon disrupts the gene's coding sequence

[Transposable elements in human genetic disease | Nature Reviews Genetics]

Transposable elements can affect genes

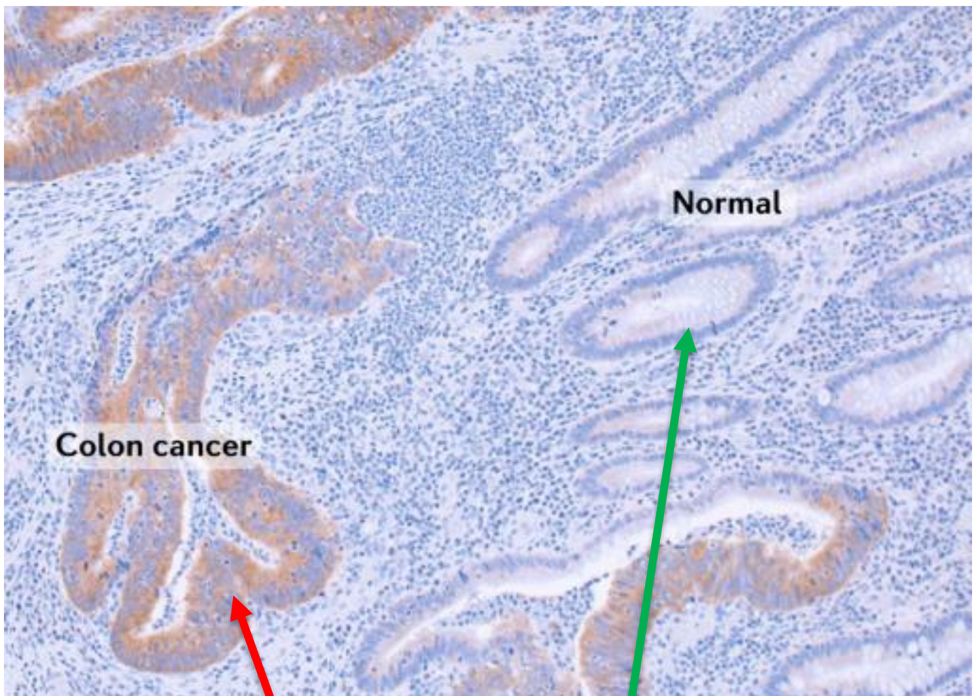
Example: Insertion of an *Alu*Ya5 element in a patient with Dent disease (a rare genetic kidney disorder) interrupts an exonic splice enhancer (ESE), resulting in skipping of exon 11, which introduces a stop codon in exon 12.



[Transposable elements in human genetic disease | Nature Reviews Genetics]

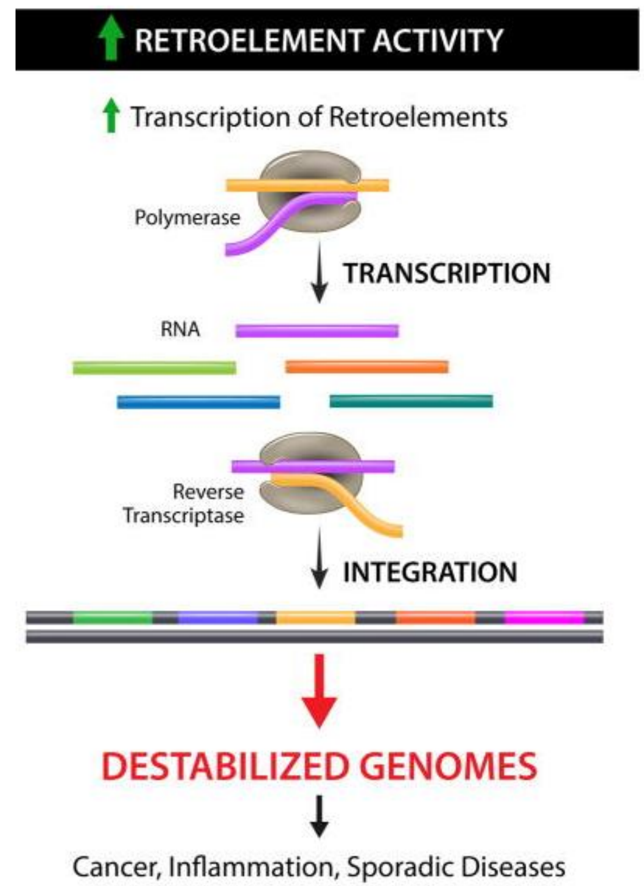
Transposable element mobility is associated with human cancers

Numerous cancers including lung, colon, pancreatic and ovarian cancers, have been associated with markers for LINE-1 mobility.



A marker for LINE-1 activity (brown)is found in cancerous tissue but not normal tissue.

HYPOTHESIS



Transposable elements in human genetic disease | Nature Reviews Genetics

Transposable elements as a source of phenotypic novelty

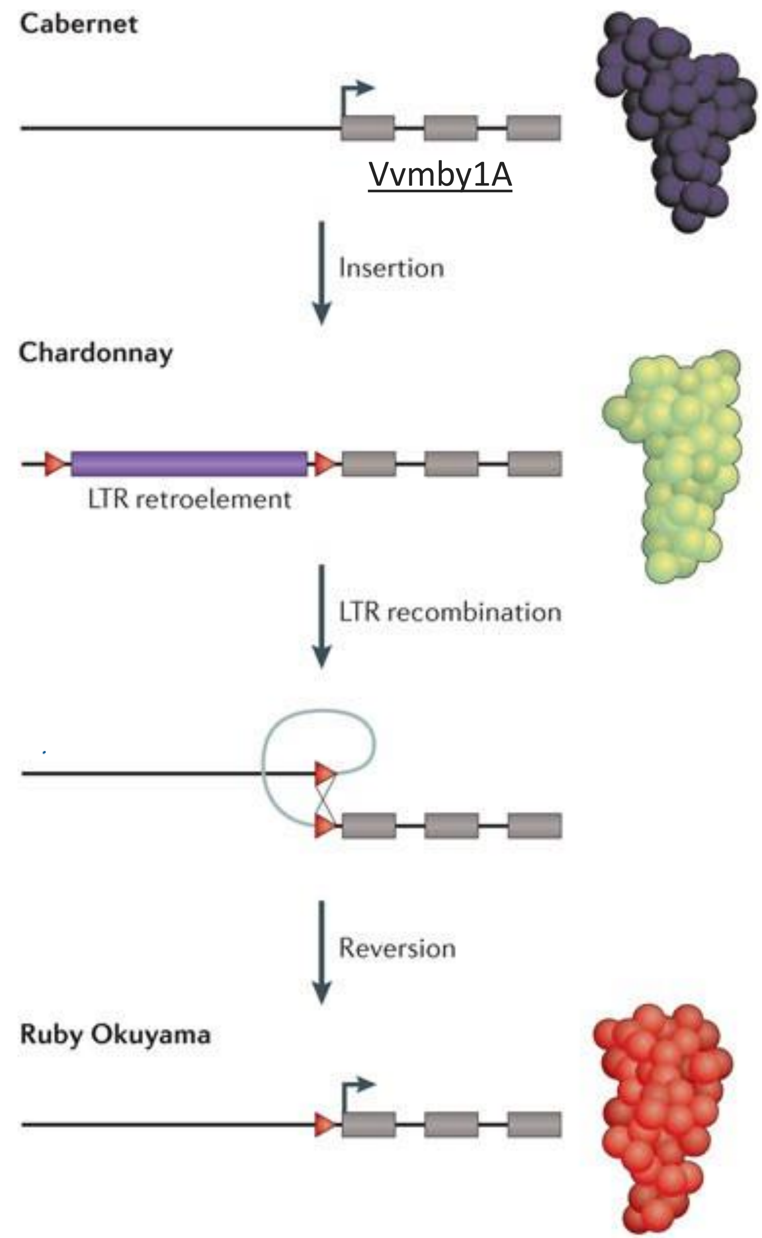
The Vvmby1A gene is required for production of purple pigment in Cabernet grapes.

An initial insertion of a *Gret1* long terminal repeat (LTR) retrotransposon resulted in a loss-of-function allele of the *Vvmby1A* gene, leading to a loss of colour in the fruit of the Chardonnay variety.

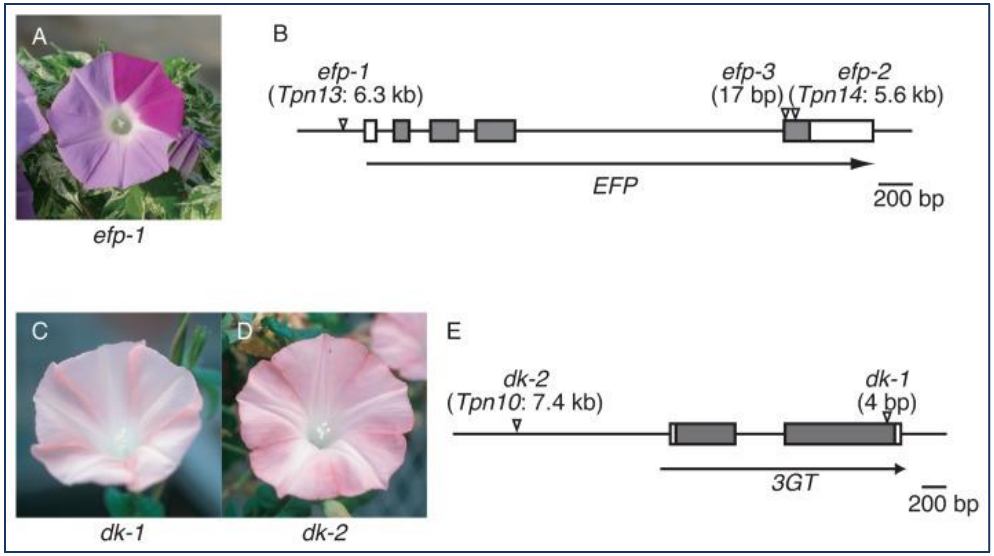
A subsequent rearrangement in *Gret1* results in revertant, coloured grapes in varieties such as Ruby Okuyama.

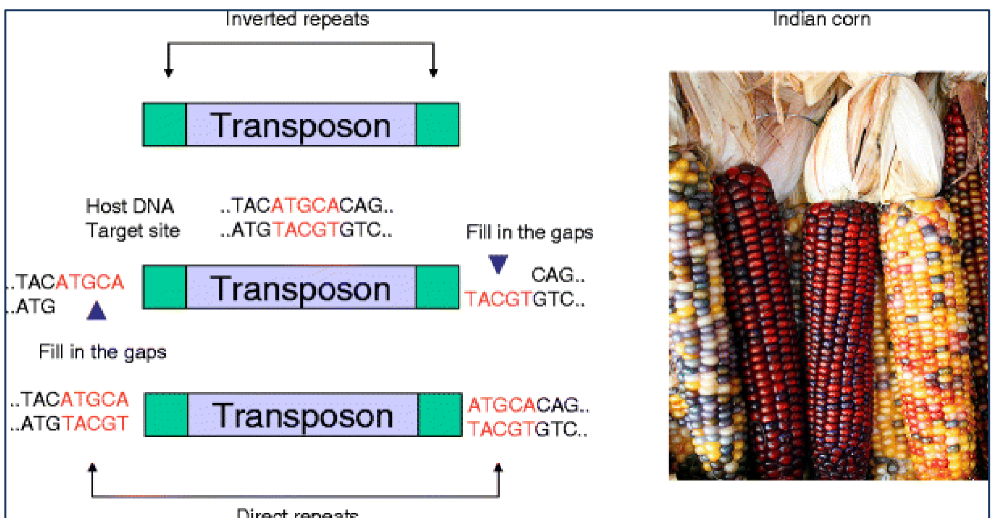
Exons are depicted as grey boxes. The LTRs flanking Gret1 just upstream of the Vvmyb1A gene are depicted as red triangles.

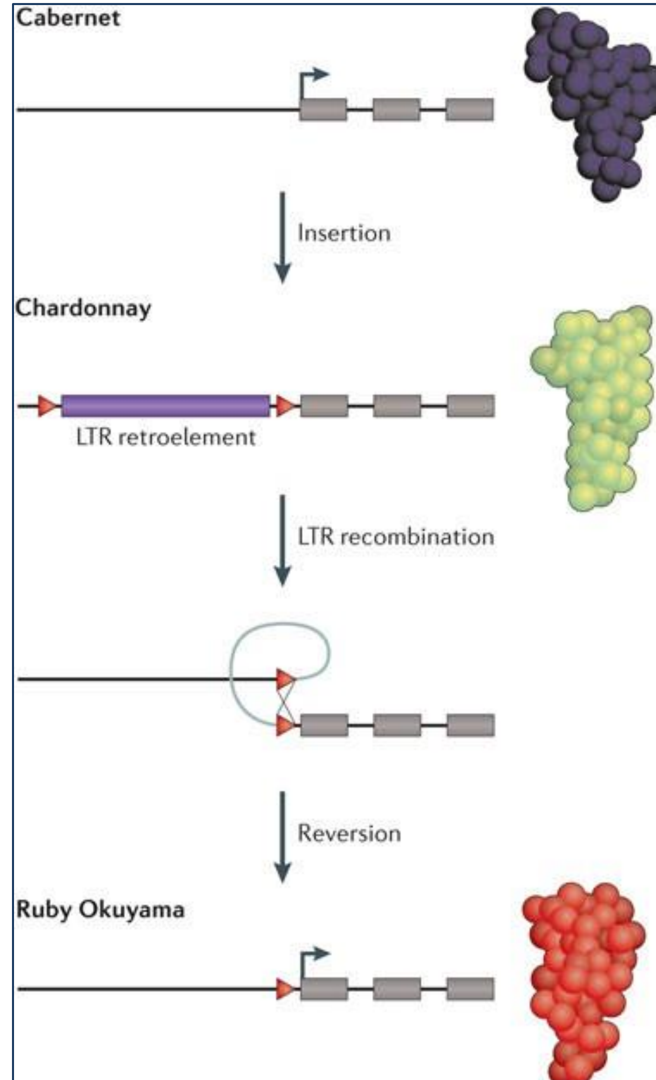
[How important are transposons for plant evolution? | Nature Reviews Genetics]



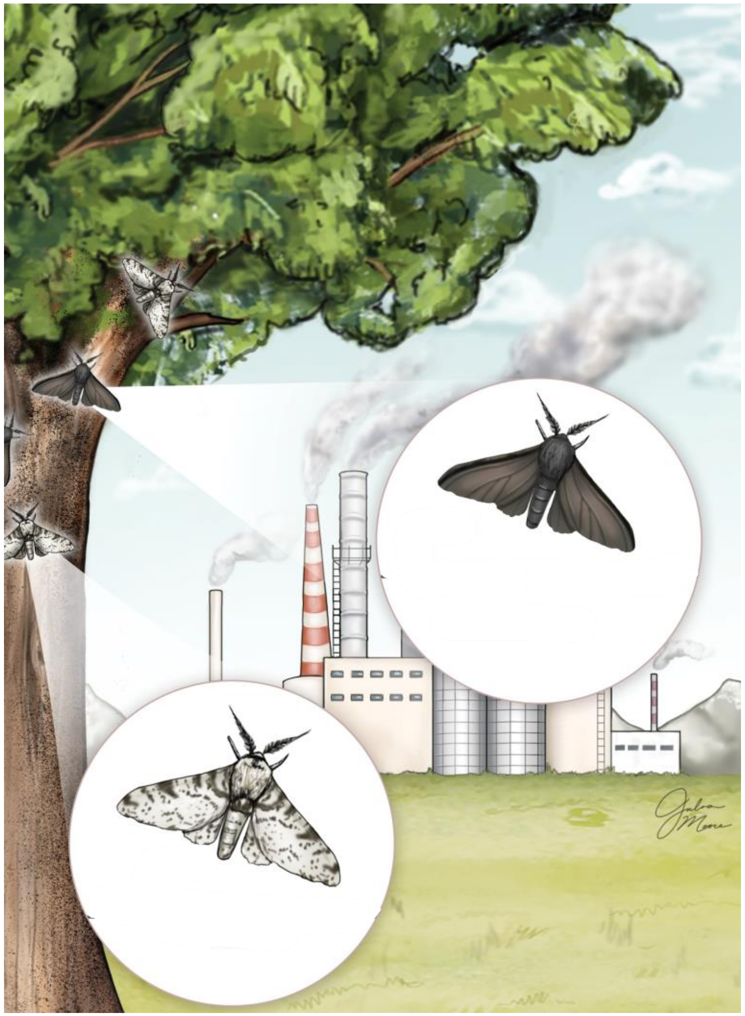
Transposable elements as a potential source of major phenotypic novelty

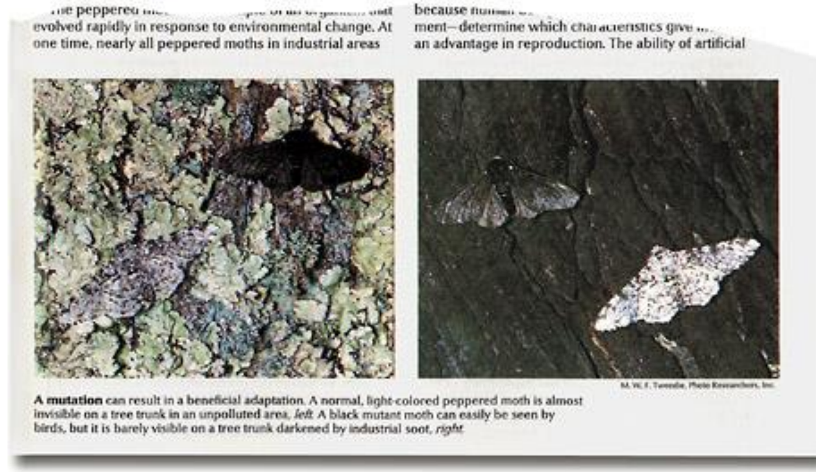






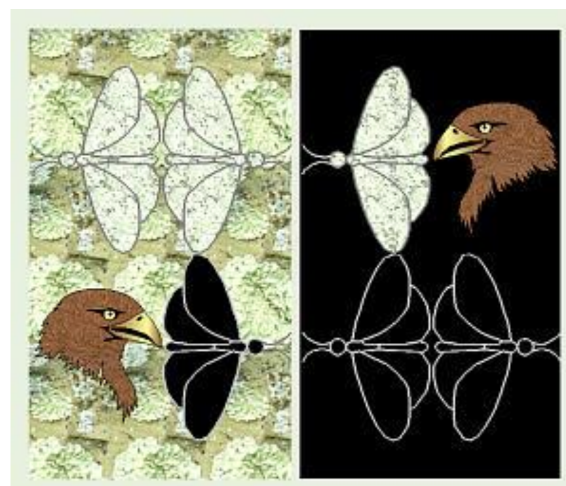
Transposable elements as a potential source of major phenotypic novelty and adaptive change




evolved rapidly in response to environmental change. At one time, nearly all peppered moths in industrial areas

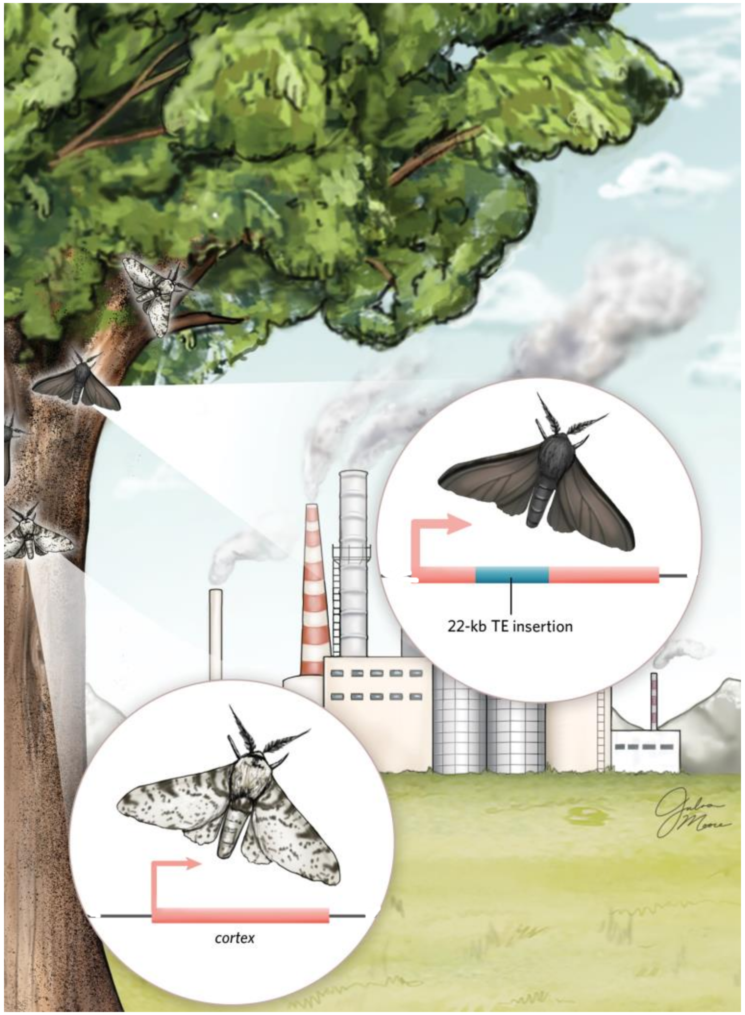
ment—determine which characteristics give … an advantage in reproduction. The ability of artificial

A mutation can result in a beneficial adaptation. A normal, light-colored peppered moth is almost invisible on a tree trunk in an unpolluted area, *left*. A black mutant moth can easily be seen by birds, but it is barely visible on a tree trunk darkened by industrial soot, *right*.



[Peppered Moths | Natural Selection Game (asu.edu)]

Transposable elements as a potential a source of major phenotypic novelty and adaptive change



(excerpt from scientific article in the journal Nature – link below)

"Here we show that the mutation event giving rise to industrial melanism in Britain was the insertion of a large, tandemly repeated, transposable element into the first intron of the gene cortex. Statistical inference based on the distribution of recombined carbonaria haplotypes indicates that this transposition event occurred around 1819, consistent with the historical record."

"The discovery that the mutation itself is a transposable element will stimulate further debate about the importance of 'jumping genes' as a source of major phenotypic novelty."

[The industrial melanism mutation in British peppered moths is a transposable element - PubMed (nih.gov)]